Linkai Peng

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TECHNICAL SKILLS

Research interest: Computer Vision, Vision Language Model, Multimodal, Medical Image Analysis
Programming: Python (Pytorch , Tensorflow/Keras, OpenCV, Scikit-Image, Sklearn, Matplotlib, Numpy)
Toolkits: CUDA, MONAI, nnUNet, SimpleITK, AWS (EC2, S3), SQL, Docker

EDUCATION

09/2023 – Present ***Northwestern University***
Ph.D. in Electrical Engineering, advisor: Prof. Ulas Bagci (https://bagcilab.com/)

09/2019 – 06/2023 ***Southern University of Science and Technology***
B.Eng in Computer Science and Technology, advisor: Prof. Xiaoying Tang

WORK EXPERIENCES

Johnson & Johnson | Research Intern Summer 2026

- Developed a **sparse video understanding** framework for endoscopic disease severity assessment, supporting both global severity classification and structured UCEIS scoring.
- Explored multimodal representations by integrating visual features with CLIP and Qwen-VL embeddings, investigating feature fusion strategies and clinically structured score prediction.

RESEARCH EXPERIENCES

Adaptation of Foundation Models 2026 - Present

- Developed a CT-based **bronchoscopy feasibility prediction** pipeline to assess airway accessibility to pulmonary nodules for clinical decision support.
- Adapted vision-language models to 3D medical imaging using LoRA for parameter-efficient fine-tuning.

Robust Medical Image Analysis 2026 - Present

- Developed **CrossPan**, a multi-center pancreas MRI benchmark (1,386 scans, 8 centers), and investigated cross-sequence generalization under varying imaging contrasts.

Multi-modal Segmentation 2024 - 2026

- Developed **TAGS**, a 3D **tumor segmentation** framework integrating SAM, CLIP-based text prompts, and anatomy-aware guidance.
- Designed a BiomedCLIP-guided **prostate lesion segmentation** framework with foreground alignment and confidence-gated refinement.

SELECTED PUBLICATIONS

2026 **Peng, L.**, et al. "Anatomy-Aware Prediction of Bronchoscopic Accessibility from 3D CT." *MICCAI 2026*.
2026 **Peng, L.**, et al. "CrossPan: A Comprehensive Benchmark for Cross-Sequence Pancreas MRI Segmentation and Generalization." *MIDL 2026*.
2025 Li, S., **Peng, L.**, et al. "TAGS: 3D Tumor-Adaptive Guidance for SAM." *ICCV-APAH 2025*.
2025 Zhang, Z., Dou, M., **Peng, L.**, et al. "VideoAds for Fast-Paced Video Understanding: Where Opensource Foundation Models Beat GPT-4o & Gemini-1.5 Pro." *ICCV 2025*.